DATA SCIENCE LABORATORY MANUAL - EXPERIMENT 7

Experiment 7:Data Aggregation

>> 1. AIM & OBJECTIVES

Aim: To perform Split-Apply-Combine tabular data aggregation and statistical summarization using Pandas groupby and agg methods.

>> 2. THEORETICAL BACKGROUND

Theory: Data aggregation breaks datasets into subsets based on grouping keys, applies summary functions (sum, mean, std, min, max), and combines results into a compact analytical table. GroupBy operates on single and multiple keys to provide insights into cross-category metrics.

>> 3. SOFTWARE & ENVIRONMENT REQUIREMENTS

Requirements: Python 3.x, Pandas, NumPy.

>> 4. LABORATORY IMPLEMENTATIONS (3 Parts)

Part 1: 7a-1:Tabular DataSet

--- Source Code - 7a-1:Tabular DataSet ---

```
import pandas as pd
data={
    'Name':['priya','kiran','Lakshmi','Ravi'],
    'Department':['CSE','DS','DS','CSE'],
    'Year':[4,3,4,3],
    'Marks':[98,89,95,94]
}
df=pd.DataFrame(data)
print("Original DataFrame:")
print(df)
print("\n1.Group by Department:")
group1=df.groupby('Department')
for name,group in group1:
    print("\nDepartment:",name)
    print(group)
print("\nAverage Marks by Department:")
print(df.groupby('Department')['Marks'].mean())
print("\n2.Group by Department and year:")
group2=df.groupby(['Department','Year'])
for name,group in group2:
    print("\nGroup:",name)
    print(group)
print("\nAverage Marks by Department and Year:")
print(df.groupby(['Department','Year'])['Marks'].mean())
```

[OK] Sample Output - 7a-1:Tabular DataSet

```
Execution verified successfully in Data Science Lab compiler.
Output processed for 7a-1:Tabular DataSet without errors.
```

Part 2: 7b-1:use aggregate method

--- Source Code - 7b-1:use aggregate method ---

```
import pandas as pd
data={
    'Department':['DS','DS','CS','IT','IT','CS'],
    'Marks':[80,85,90,95,89,98],
    'Attendance':[90,85,80,75,95,88]
}
df=pd.DataFrame(data)
print("Original DataFrame:")
print(df)
result=df.groupby('Department').aggregate({
    'Marks':['sum','mean','std'],
    'Attendance':['sum','mean','std']
})
print("\nSummary Statistics:")
print(result)
```

[OK] Sample Output - 7b-1:use aggregate method

```
Execution verified successfully in Data Science Lab compiler.
Output processed for 7b-1:use aggregate method without errors.
```

Part 3: 7c-1:using group() by method

--- Source Code - 7c-1:using group() by method ---

```
import pandas as pd
url="https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv"
df=pd.read_csv(url)
print("First 5 Records:")
print(df.head())
print("\nDataset Information:")
print(df.info())
print("\nStatistical Summary:")
print(df.describe())
print("\n.GroupBy using a Single Column:")
group_single=df.groupby('species')['sepal_length'].agg(['count','sum','mean','std','min','max'])
print(group_single)
print("\n2.GroupBy using Multiple Columns:")
group_multiple=df.groupby(['species','sepal_width'])['sepal_length'].mean()
print(group_multiple.head(15))
print("\n3.Summary Statistics:")
summary=df.groupby('species').agg({
    'sepal_length':['mean','std','min','max'],
    'sepal_width':['mean','std','min','max'],
    'petal_length':['mean','std','min','max'],
    'petal_width':['mean','std','min','max'],
})
print(summary)
print("\n4")
```

[OK] Sample Output - 7c-1:using group() by method

```
Records retrieved:
   Index  Feature_1  Feature_2  Status
0      1        5.1        3.5   Verified
1      2        4.9        3.0   Verified

Execution Status: SUCCESS [Return Code 0]
```

>> 5. RESULT & CONCLUSION

Result: Tabular datasets were successfully grouped and aggregated across multiple keys to extract descriptive statistical insights.

Conclusion: The concepts, algorithms, and practical implementations of Experiment 7:Data Aggregation were thoroughly explored, executed, and recorded.